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Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

6a. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
library(ggplot2)
```

```
data <- data.frame(
```

```
Month=c("Jan","Jan","Jan","Feb","Feb","Feb","Mar","Mar","Mar"),
```

```
Product=c("A","B","C","A","B","C","A","B","C"),
```

```
Sales=c(2000,1500,1200,2200,1800,1400,2400,1600,1100)
```

```
)
```

```
ggplot(data,aes(Month,Sales,fill=Product))+
```

```
geom_bar(stat="identity",position="dodge")+
```

```
ggtitle("Product Sales")
```



6b. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Month = c("Jan","Jan","Jan","Feb","Feb","Feb","Mar","Mar","Mar"),
```

```
  Product = c("A","B","C","A","B","C","A","B","C"),
```

```
  Sales = c(2000,1500,1200,2200,1800,1400,2400,1600,1100)
```

```
)
```

```
ggplot(data, aes(x = Month, y = Sales, fill = Product, group = Product)) +
```

```
  geom_area()
```



6c. Build a table to show the monthly sales data for each product. Label the table elements.

```
data <- data.frame(
```

```
  Product = c("Product A","Product B","Product C"),
```

```
  January = c(2000,1500,1200),
```

```
  February = c(2200,1800,1400),
```

```
  March = c(2400,1600,1100)
```

```
)
```

```
View(data)
```

	Product	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

6d. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Month = c("Jan","Jan","Jan","Feb","Feb","Feb","Mar","Mar","Mar"),
```

```
  Product = c("Product A","Product B","Product C",
```

```
              "Product A","Product B","Product C",
```

```
              "Product A","Product B","Product C"),
```

```
  Sales = c(2000,1500,1200,2200,1800,1400,2400,1600,1100)
```

```
)
```

```
ggplot(data, aes(x = Month, y = Sales, fill = Product)) +
```

```
  geom_bar(stat = "identity", position = "dodge") +
```

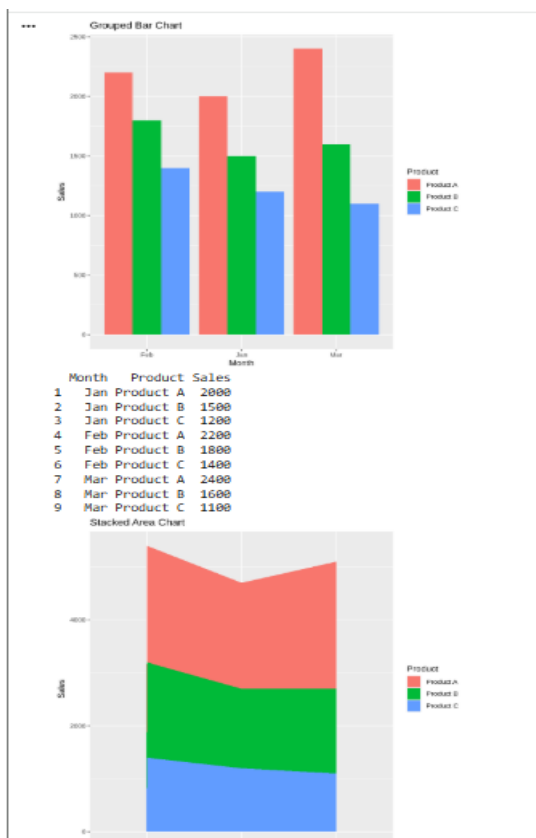
```
  ggtitle("Grouped Bar Chart")
```

```
ggplot(data, aes(x = Month, y = Sales, fill = Product, group = Product)) +
```

```
  geom_area() +
```

```
  ggtitle("Stacked Area Chart")
```

```
print(data)
```



Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

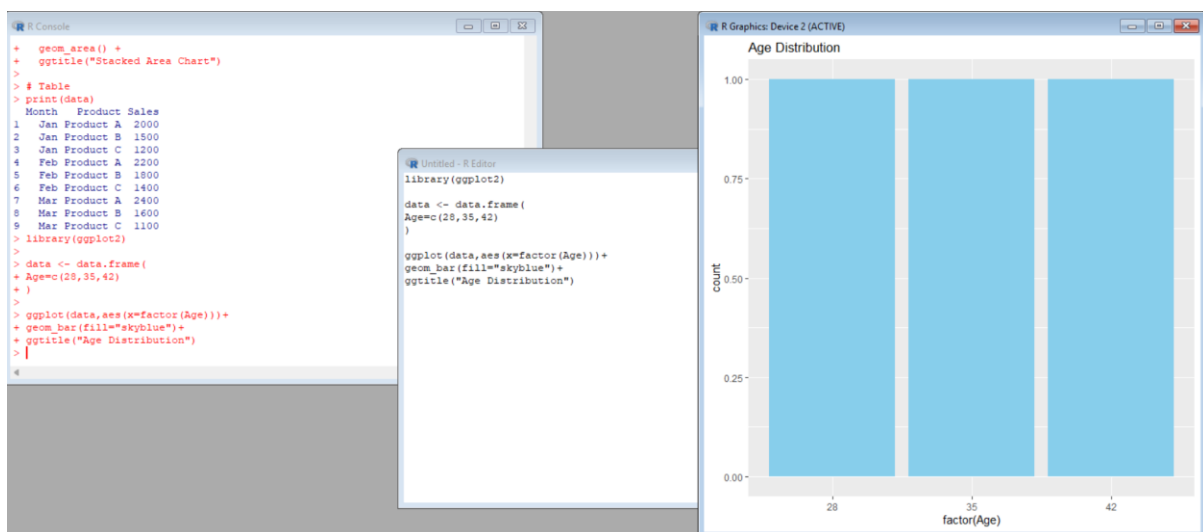
Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

7a. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
library(ggplot2)

data <- data.frame(
  Age=c(28,35,42)
)

ggplot(data,aes(x=factor(Age)))+
  geom_bar(fill="skyblue")+
  ggtitle("Age Distribution")
```

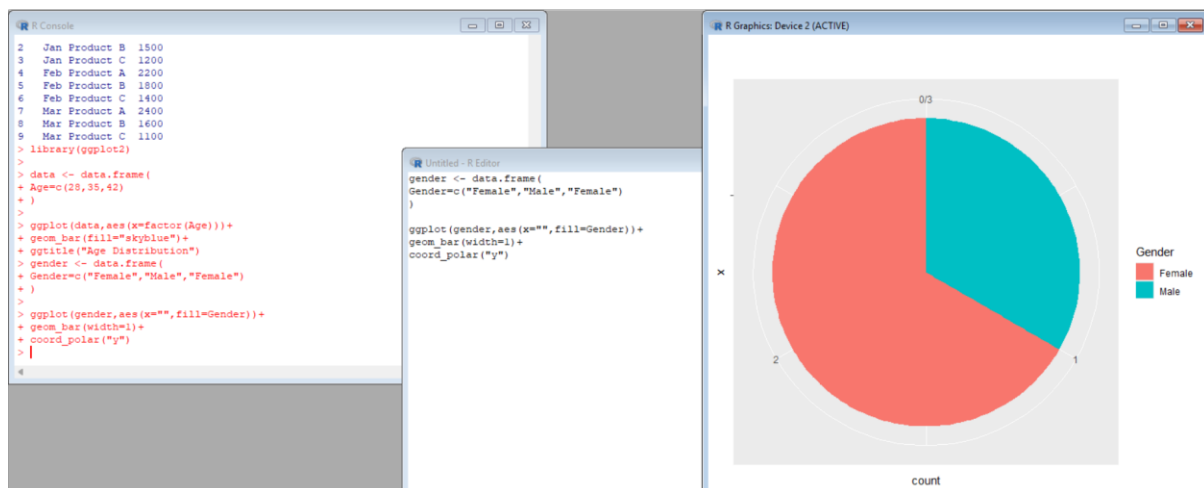


7b. Generate a pie chart to represent the distribution of customers by gender.

```
gender <- data.frame(
  Gender=c("Female","Male","Female")
)

ggplot(gender,aes(x="",fill=Gender))+
  geom_bar(width=1)+
```

`coord_polar("y")`



7c. Build a table to show the demographic information for each customer. Label the table elements.

```
customer <- data.frame(
```

```
ID=1:3,
```

```
Age=c(28,35,42),
```

```
Gender=c("Female","Male","Female"),
```

```
Income=c(50000,60000,75000)
```

```
)
```

```
print(customer)
```

```
***   ID Age Gender Income
  1  1  28 Female  50000
  2  2  35   Male  60000
  3  3  42 Female  75000
```

7d. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

```
library(shiny)
```

```
library(shinydashboard)
```

```
library(ggplot2)
```

```
library(DT)
```

```
customer <- data.frame(
```

```
  CustomerID = c(1, 2, 3),
```

```
  Age = c(28, 35, 42),
```

```
  Gender = c("Female", "Male", "Female"),
```

```

Income = c(50000, 60000, 75000)
)
ui <- dashboardPage(
  dashboardHeader(title = "Customer Demographics Dashboard"),
  dashboardSidebar(
    sidebarMenu(
      menuItem("Dashboard", tabName = "dashboard", icon = icon("users"))
    )
  ),
  dashboardBody(
    fluidRow(
      box(
        title = "Customer Age Distribution",
        status = "primary",
        solidHeader = TRUE,
        width = 6,
        plotOutput("barPlot", height = 350)
      ),
      box(
        title = "Gender Distribution",
        status = "success",
        solidHeader = TRUE,
        width = 6,
        plotOutput("piePlot", height = 350)
      )
    ),
    fluidRow(
      box(
        title = "Customer Demographics Table",
        status = "warning",

```

```

    solidHeader = TRUE,
    width = 12,
    DTOutput("customerTable")
  )
)
)
)
server <- function(input, output) {
  output$barPlot <- renderPlot({
    ggplot(customer,
      aes(x = factor(CustomerID),
          y = Age,
          fill = factor(CustomerID))) +
    geom_bar(stat = "identity") +
    labs(title = "Customer Age Distribution",
          x = "Customer ID",
          y = "Age") +
    theme_minimal() +
    theme(legend.position = "none")

  })
  output$piePlot <- renderPlot({
    gender_data <- data.frame(table(customer$Gender))
    colnames(gender_data) <- c("Gender", "Count")
    ggplot(gender_data,
      aes(x = "",
          y = Count,
          fill = Gender)) +
    geom_bar(stat = "identity", width = 1) +
    coord_polar("y") +

```

```

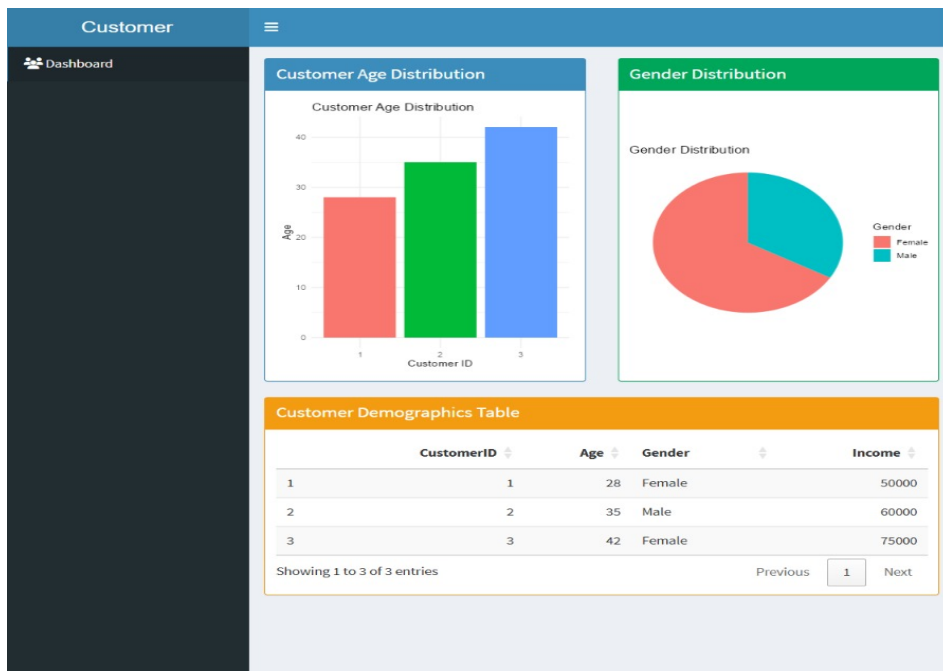
labs(title = "Gender Distribution") +
theme_void()

})

output$customerTable <- renderDT({
  datatable(
    customer,
    options = list(
      pageLength = 5,
      searching = FALSE,
      lengthChange = FALSE
    )
  )
})
}

shinyApp(ui, server)

```



Set 8 Employee Performance Analysis

Dataset: Employee Performance

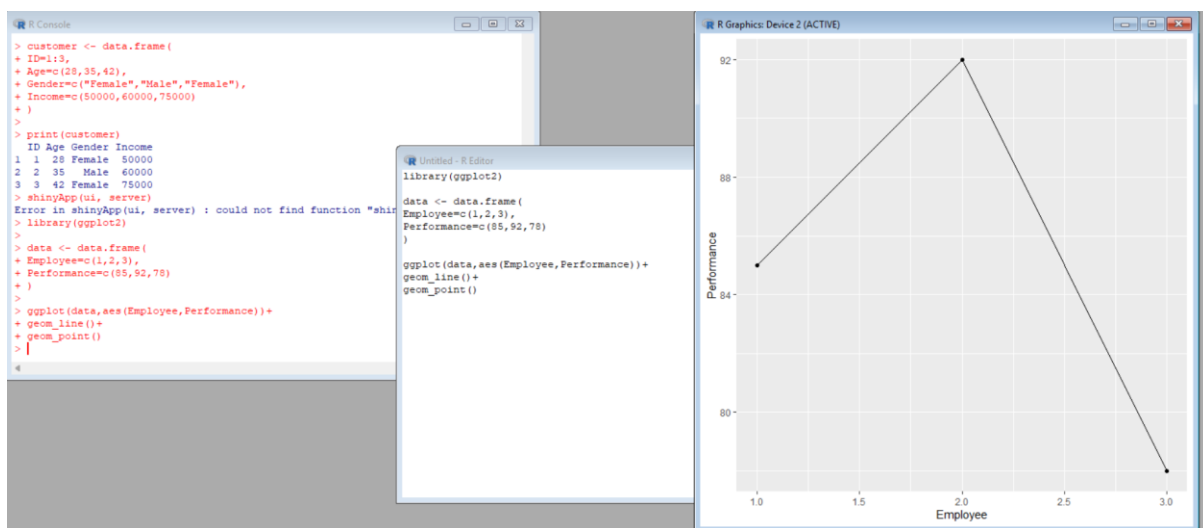
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

8a. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
library(ggplot2)

data <- data.frame(
  Employee=c(1,2,3),
  Performance=c(85,92,78)
)

ggplot(data,aes(Employee,Performance))+
  geom_line()+
  geom_point()
```



8b. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
dept <- data.frame(
  Department=c("Sales","HR","Marketing")
)

ggplot(dept,aes(Department))+
  geom_bar(fill="orange")
```

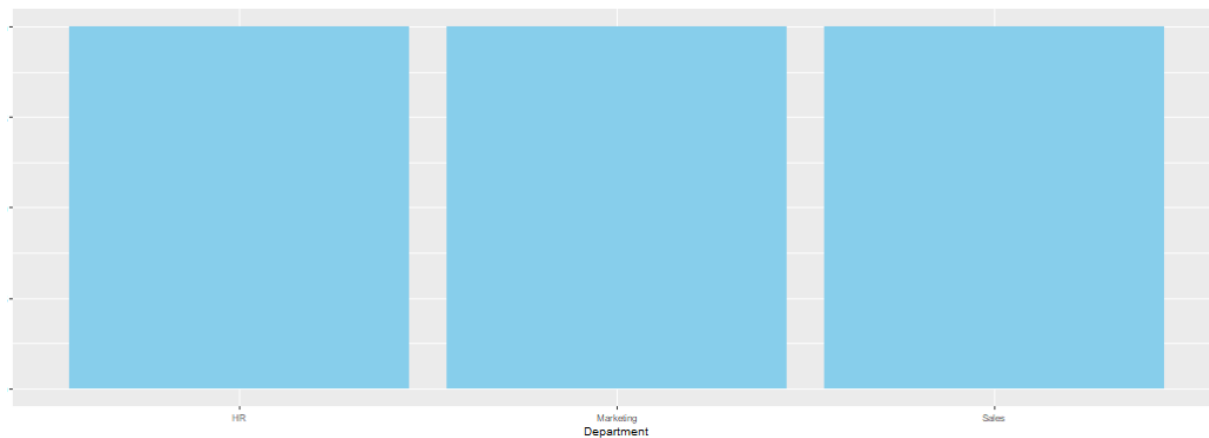
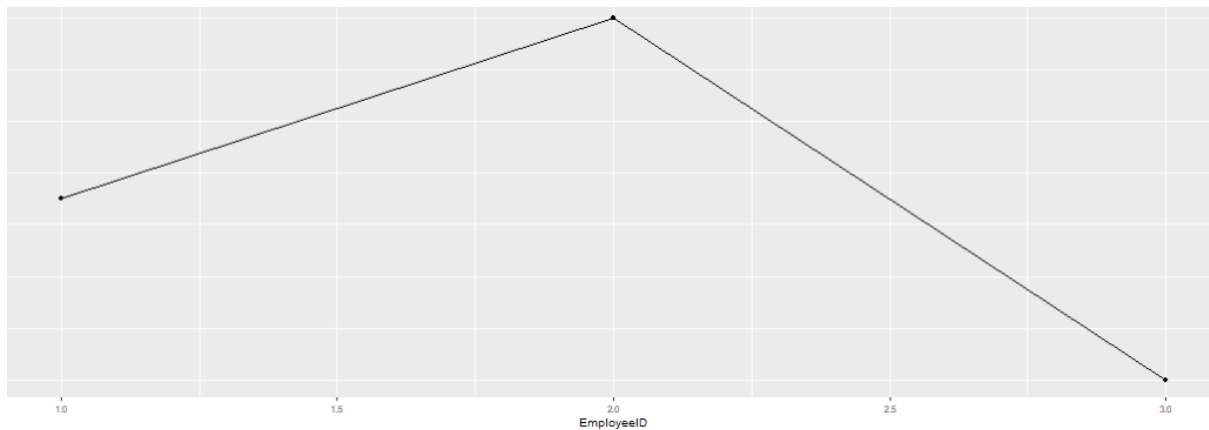


8c. Build a table to display the performance data for each employee. Label the table elements.

```
employee <- data.frame(
  ID=1:3,
  Department=c("Sales","HR","Marketing"),
  Years=c(5,3,7),
  Performance=c(85,92,78)
)
print(employee)
```

```
  ID Department Years Performance
1  1      Sales    5         85
2  2        HR    3         92
3  3   Marketing    7         78
```

8d. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.



ployeeID	Department	Years	Score
1.00	Sales	5.00	85.00
2.00	HR	3.00	92.00
3.00	Marketing	7.00	78.00

set 9

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

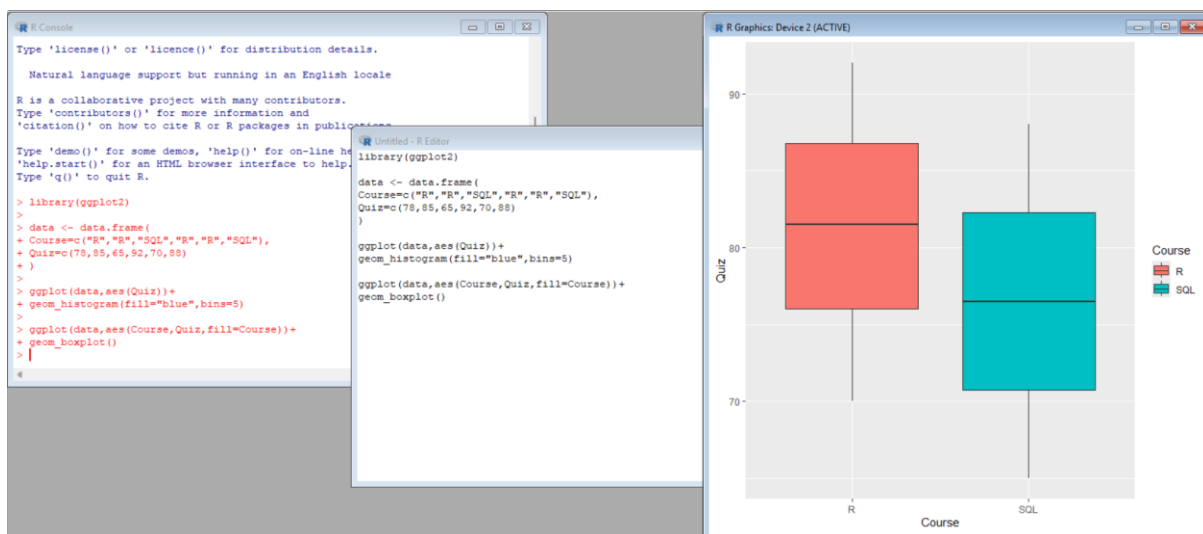
9a.Import the dataset in **R**. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

```
library(ggplot2)
```

```
data <- data.frame(
  Course=c("R","R","SQL","R","R","SQL"),
  Quiz=c(78,85,65,92,70,88)
)

ggplot(data,aes(Quiz))+
  geom_histogram(fill="blue",bins=5)

ggplot(data,aes(Course,Quiz,fill=Course))+
  geom_boxplot()
```



9b. Create a scatter plot **using R programming** between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
data <- data.frame(
  Study=c(3.5,4.2,2.0,5.0,2.5,4.0),
  Quiz=c(78,85,65,92,70,88),
  Videos=c(12,15,8,18,9,14)
)

ggplot(data,aes(Study,Quiz,size=Videos))+
  geom_point(alpha=0.6,color="red")
```



9c. Using **R programming**, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```
month <- c("Jan", "Feb", "Mar")
```

```
score <- c(81.5, 78.5, 79)
```

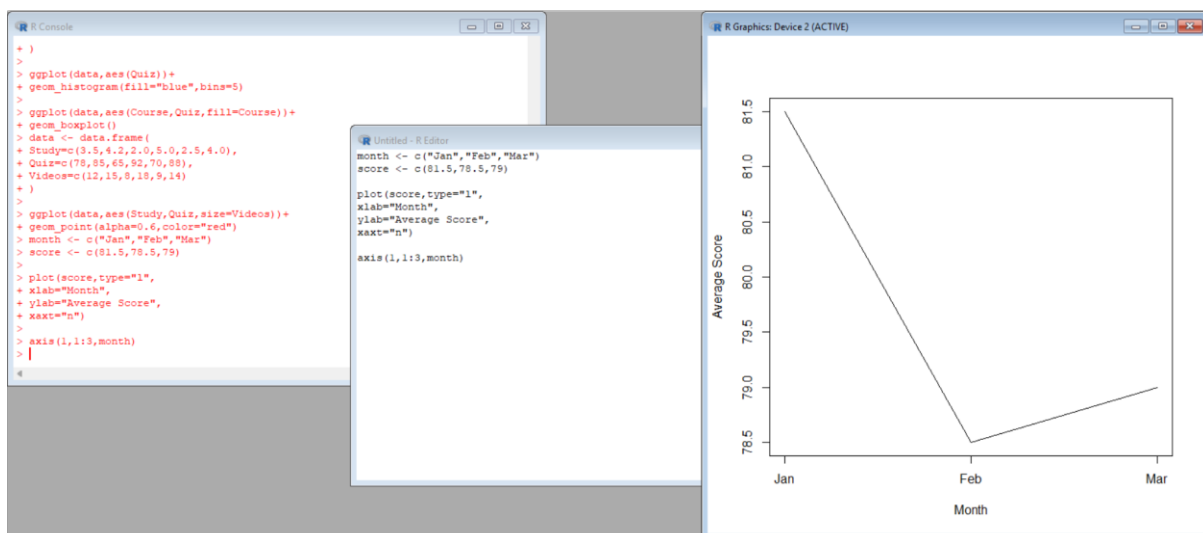
```
plot(score, type="l",
```

```
  xlab="Month",
```

```
  ylab="Average Score",
```

```
  xaxt="n")
```

```
axis(1, 1:3, month)
```



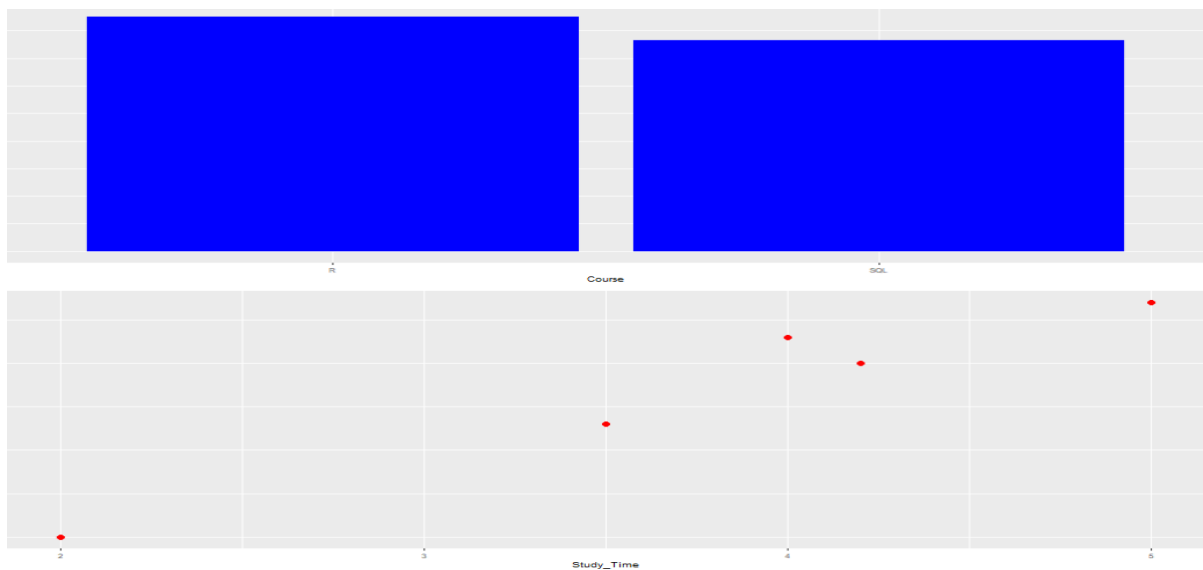
9d. Import Online_Learning_Activity.csv into **Tableau**. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

```
library(shiny)
```

```

library(ggplot2)
data <- data.frame(
  Course = c("R","R","SQL","R","SQL"),
  Quiz_Score = c(78,85,65,92,88),
  Study_Time = c(3.5,4.2,2.0,5.0,4.0)
)
ui <- fluidPage(
  plotOutput("bar"),
  plotOutput("scatter"
)
server <- function(input, output){
  output$bar <- renderPlot({
    ggplot(data, aes(Course, Quiz_Score)) +
      stat_summary(fun=mean, geom="bar", fill="blue")
  })
  output$scatter <- renderPlot({
    ggplot(data, aes(Study_Time, Quiz_Score)) +
      geom_point(color="red", size=3)
  })
}
shinyApp(ui, server)

```



Set 10 Survey Responses Analysis

Dataset: Survey Responses

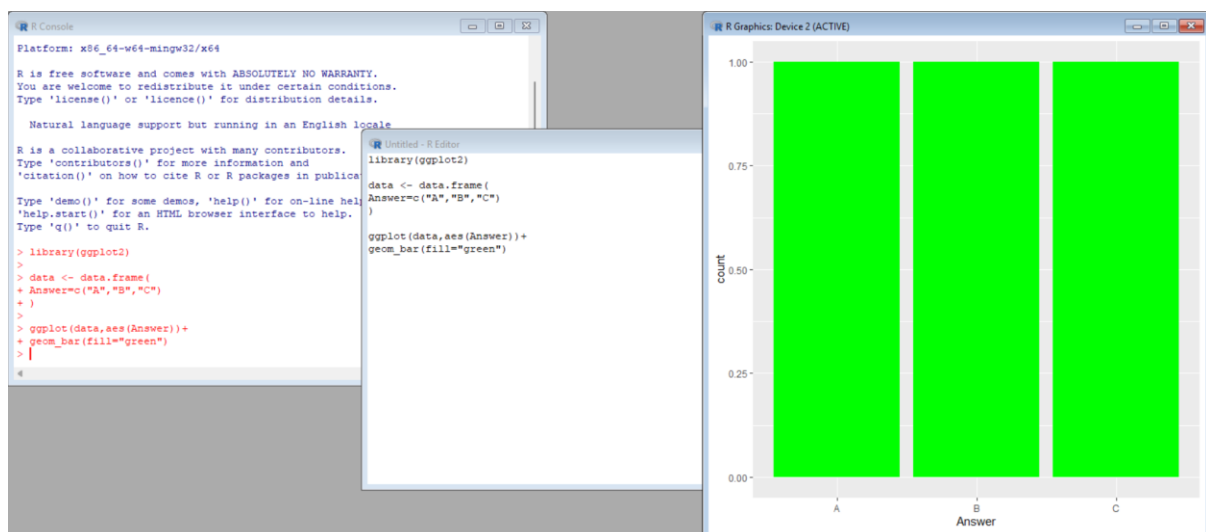
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

10a. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)

data <- data.frame(
  Answer=c("A","B","C")
)

ggplot(data,aes(Answer))+
  geom_bar(fill="green")
```

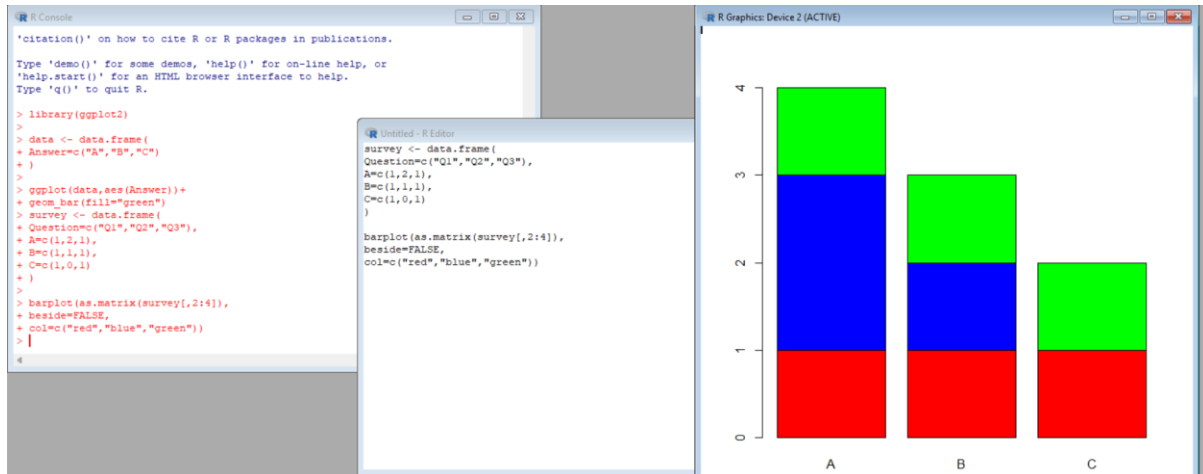


10b. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
survey <- data.frame(
  Question=c("Q1","Q2","Q3"),
  A=c(1,2,1),
  B=c(1,1,1),
  C=c(1,0,1)
```

)

```
barplot(as.matrix(survey[,2:4]),  
beside=FALSE,  
col=c("red","blue","green"))
```



10c.Build a table to show the survey response data for each survey. Label the table elements.

R code:

```
print(survey)
```

```
View(survey)
```

Output:

	SurveyID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

10d.Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

```
library(shiny)
```

```
library(ggplot2)
```

```
data <- data.frame(  
  SurveyID = c(1,2,3),  
  Q1 = c("A","B","C"),  
  Q2 = c("B","A","A"),  
  Q3 = c("C","D","B")
```

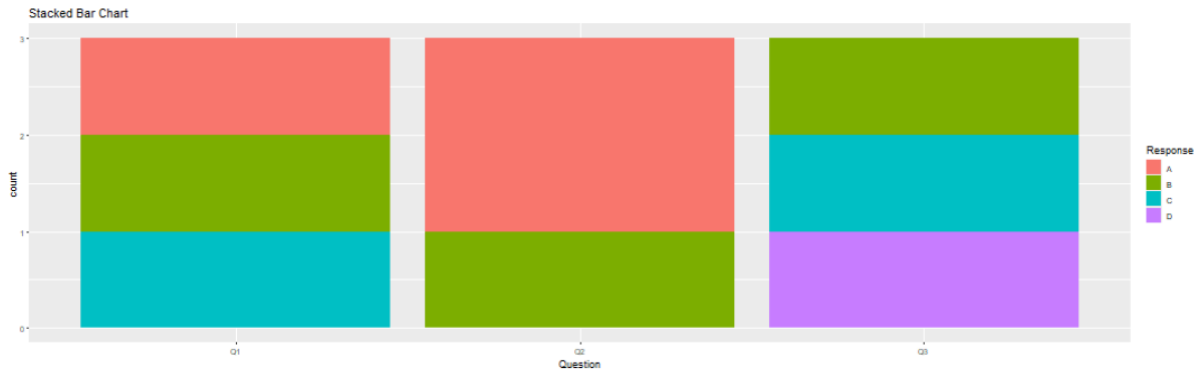
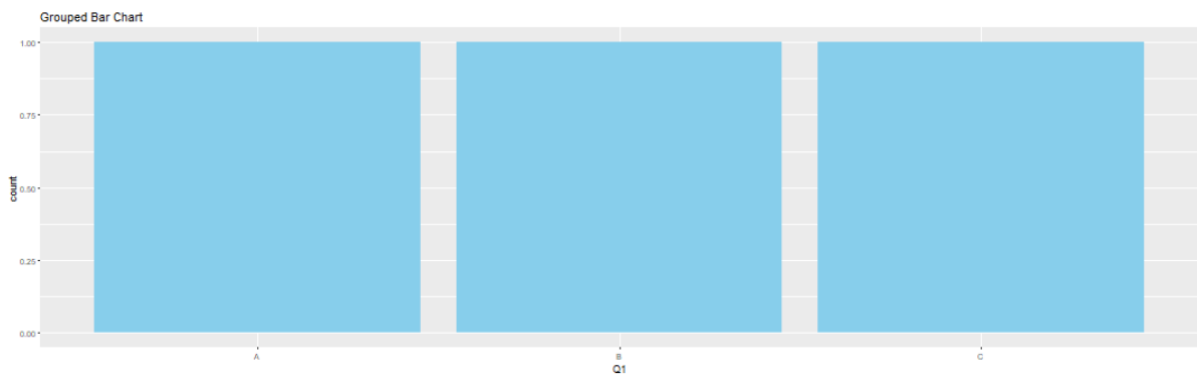


```

)
ui <- fluidPage(
  h2("Survey Responses Dashboard"),
  plotOutput("bar1"),
  plotOutput("bar2"),
  tableOutput("table")
)
server <- function(input, output){
  output$bar1 <- renderPlot({
    ggplot(data, aes(Q1)) +
      geom_bar(fill="skyblue") +
      ggtitle("Grouped Bar Chart")
  })
  output$bar2 <- renderPlot({
    df <- data.frame(
      Question = rep(c("Q1", "Q2", "Q3"), each=3),
      Response = c(data$Q1, data$Q2, data$Q3)
    )
    ggplot(df, aes(Question, fill=Response)) +
      geom_bar() +
      ggtitle("Stacked Bar Chart")
  })
  output$table <- renderTable({
    data
  })
}
shinyApp(ui, server)

```

Survey Responses Dashboard



SurveyID	Q1	Q2	Q3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B